| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 04/30/2017 |
| **PI Name:** | Limoli, Charles Ph.D. | | |
| **Project Title:** | NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | **TechPort:** | No | |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | climoli@uci.edu | **Fax:** | FY (949) 824-3566 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | (949) 824-3053 |
| **Organization Name:** | University of California | | |
| **PI Address 1:** | Dept. of Radiation Oncology | | |
| **PI Address 2:** | Medical Sciences I, B149 | | |
| **PI Web Page:** | | | |
| **City:** | Irvine | **State:** | CA |
| **Zip Code:** | 92697-2695 | **Congressional District:** | 45 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2013-14 HERO NNJ13ZSA002N-NSCOR Radiation |
| **Start Date:** | 07/01/2015 | **End Date:** | 06/30/2020 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 4 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 5 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Simonsen, Lisa | **Contact Phone:** | |
| **Contact Email:** | lisa.c.simonsen@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Acharya, Munjal Ph.D. ( University of California, Irvine )<br>Baulch, Janet Ph.D. ( University of California, Irvine )<br>Britten, Richard Ph.D. ( Eastern Virginia Medical School )<br>Nelson, Gregory Ph.D. ( Loma Linda University )<br>Parihar, Vipan Ph.D. ( University of California, Irvine )<br>Soltesz, Ivan Ph.D. ( University of California, Irvine )<br>Vlkolinsky, Roman Ph.D. ( Loma Linda University ) | | |
| **Grant/Contract No.:** | NNX15AI22G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining | | |

| | |
|---|---|
| | whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources. |
| **Task Progress:** | Overview of progress and administrative matters:<br>The overall organization of the CNS NASA Specialized Center of Research (NSCOR) remains unchanged, where it is managed at the University of California, Irvine (UCI: PI & Director, Limoli) and is conducted between 3 other universities, including Stanford (PI and Co-director, Soltesz), Loma Linda University (LLU: PIs Nelson & Vlkolinsky), and Eastern Virginia Medical School (EVMS: PI, Britten). The overarching hypothesis that links these studies seeks to define the mechanisms contributing to charged particle–induced disruptions to CNS functionality that impact risk over acute (mission critical) and more chronic (terrestrial) timeframes. The unique composition of the NSCOR team has begun to deliver some significant and unexpected findings (highlighted below), and is poised to embark on some exciting new studies implementing GCR (galactic cosmic radiation) simulation and low dose rate neutron studies to further define specific and highly relevant space radiation effects in the CNS. Our studies have now begun to shed considerable light regarding radiation effects pathway from the synapse to multiple neurocognitive complications. Work has uncovered some truly remarkable and persistent effects spanning a 1-year timeframe in which behavioral decrements are temporally coincident with long-range disruptions in circuit connectivity across multiple brain regions, changes transpiring amidst significant increases in neuroinflammation.<br><br>These studies are being coordinated with field recordings of neuronal ensembles within distinct cortical regions, and with studies focused on the role of activated microglia and secreted exosomes in regulating the radiation response of the brain. Additional work has been instrumental in defining a key role for epigenetics, and how radioresponsive changes in DNA methylation and hydroxymethylation control the expression of key genes known to impact learning and memory. Many of the foregoing studies have been conducted in both mice and rats utilizing male and female animals exposed at astronaut relevant ages (6 months for mice, 9 months for rats).<br><br>Research Highlights for the CNS NSCOR:<br><br>Work under the NSCOR has been able to demonstrate direct links between charged particle induced disruptions to cognition and persistent damage to specific neuronal subsets interrogated by our behavioral tasks. We have now published 2 seminal papers that have detailed our findings linking behavioral changes to structural plasticity in the irradiated brain. The first was published in Science Advances and the second published in Scientific Reports.<br><br>1. Parihar, V.K., Allen, B., Tran, K.K., Macaraeg, T., Chu, E., Kwok, S., Chmielewski, N.N., Craver, B.M., Baulch, J.E., Acharya, M.M., Cucinotta, F.A. and Limoli, C.L. What happens to your brain on the way to Mars. Science Advances, May 1;1(4): e1400256. <a target="_blank" href="http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=pubmed&dopt=Abstract&list_uids=26180843">PMID: 26180843</a> , (2015).<br><br>2. Parihar, V.K., Allen, B.D., Caressi, C., Kwok, S., Chu, E., Tran, K.K., Chmielewski, N.N., Giedzinski, E., Acharya, M.M., Britten, R.A., Baulch J.E. and Limoli, C.L. Cosmic radiation exposure and persistent cognitive dysfunction Sci. Rep. 10;6:34774. doi:10.1038/srep34774. <a target="_blank" href="http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=pubmed&dopt=Abstract&list_uids=27721383">PMID:27721383</ (2016).<br><br>Each of these papers received significant worldwide press, with the former being picked up by 52 news agencies, ranking 5th in altmetrics and 7th in total page views, while the latter was picked up by over 70 news agencies and made the top 100 cited papers for that journal during the entire calendar year. These findings prompted several popular magazines to publish synopsis of our work (e.g., National Geographic, Smithsonian, USA today), including a recent review published by the PI in Scientific American.<br><br>3. Limoli, C.L. Deep-Space Deal Breaker. Scientific American, 316(2):54-59 (2017).<br><br>Radiation-induced impairments in spatial, episodic, and recognition memory were temporally coincident with deficits in executive function and reduced rates of fear extinction and elevated anxiety. Irradiation caused significant reductions in dendritic complexity, spine density, and altered spine morphology along medial prefrontal cortical neurons known to mediate neurotransmission interrogated by our behavioral tasks. Each of these space radiation-induced changes was found to persist out to 6 months following exposure.<br><br>We have also made some significant insights into how charged particle exposure impacts epigenetics by elevating the levels of 5-methyl- and 5-hydroxymethylcytosine. Increased levels of these DNA modifications are associated with poor behavioral performance, and pharmacologic inhibition of these epigenetic changes have now been shown to resolve radiation-induced cognitive deficits, laying the groundwork for further neuroepigenetic studies. These seminal findings have now been published in Scientific Reports.<br><br>4. Acharya MM, Baddour AA, Kawashita T, Allen BD, Syage, AR, Nguyen TH, Yoon N, Giedzinski E, Yu L, Parihar VK, Baulch JE (2017) Epigenetic determinants of space radiation-induced cognitive dysfunction. Sci Rep. 2017 Feb 21;7:42885. Doi: 10.1038/srep42885.<br><br>Electrophysiological studies have uncovered some fascinating effects of cosmic radiation exposure of within the hippocampus and other cortical regions of the brain. In one study, GABA release from cannabinoid type 1 receptor (CB1)-expressing basket cells (CB1BCs) onto pyramidal cells (PCs) was dramatically increased. Since this effect was abolished by CB1 blockade, data indicates that irradiation decreased CB1-dependent tonic inhibition of GABA release. Such findings reveal a surprising selective long-term plasticity of synaptic microcircuits in the hippocampus.<br><br>5. Lee, S.-L., Dudok, B., Parihar, V.K., Jung, K.-M., Zöldi, M., Kang, Y.-J., Maroso, M., Alexander, A.L., Nelson, G.A., Piomelli, D., Katona, I., Limoli, C.L. & Soltesz, I. Neurophysiology of space travel: Energetic solar particles cause cell type-specific plasticity of neurotransmission. Brain Struct. Func. (2017) In Press. DOI:10.1007/s00429-016-1345-3<br><br>Interestingly, such unanticipated effects of cosmic radiation exposure are not limited to microcircuits, but extend to longer range disruptions in neurotransmission between distinct regions of the brain. Studies conducted over the course of 1 year now show that connectivity between the hippocampus and perirhinal cortex are virtually abolished, revealing a remarkably persistent deficit that increases the neurocognitive risks associated with deep space travel. These results are currently under review at:<br><br>6. Parihar, V.K., Maroso, M., Syage, A., Allen, B.D., Angulo, M.C., Soltesz, I. and Limoli, C.L. Cosmic radiation short circuits neurotransmission amid elevated inflammation to disrupt cognition 1 year after exposure. Submitted Scientific Reports. (2017).<br><br>Work with rats between EVMS and LLU is progressing nicely and starting to reveal some interesting effects. A significant number of rats prescreened for Attentional Set Shifting (ATSET) performance were irradiated at Brookhaven National Laboratory (BNL) and sent to LLU for electrophysiological assessments. Much of this work is currently underway. Some of the initial findings have clearly indicated significant impairment of Simple Discrimination (SD) performance, with considerable inter-individual variability in performance.<br><br>Work at LLU is presently exploring the electrophysiological properties of rat cortical regions in these behaviorally trained rats following |

| | |
|---|---|
| | silicon radiation exposure.<br><br>In summary, much of our findings in this initial phase of the NSCOR funding period have now laid the groundwork for some exciting upcoming studies utilizing the GCR simulation at BNL and some low dose rate studies with neutrons at Colorado State University. Our findings have now provided an ideal segue into our new data sets that significantly expand our prior knowledge concerning the mechanisms of how cosmic radiation damages the brain and compromises neurotransmission. Our work sheds light on many of the disrupted processes in the irradiated brain that we posit will significantly elevate the risk for manifesting early and late CNS dysfunction during and after deep space travel. |
| **Bibliography Type:** | Description: (Last Updated: 01/14/2026) |
| **Articles in Other Journals or Periodicals** | Limoli CL. "Deep-space deal breaker." Scientific American. 2017 Jan 17;316(2):54-9. http://dx.doi.org/10.1038/scientificamerican0217-54 ; PubMed PMID: 28118333 , Jan-2017 |
| **Articles in Peer-reviewed Journals** | Parihar VK, Allen BD, Caressi C, Kwok S, Chu E, Tran KK, Chmielewski NN, Giedzinski E, Acharya MM, Britten RA, Baulch JE, Limoli CL. "Cosmic radiation exposure and persistent cognitive dysfunction." Sci Rep. 2016 Oct 10;6:34774. http://dx.doi.org/10.1038/srep34774 ; PubMed PMID: 27721383; PubMed Central PMCID: PMC5056393 , Oct-2016 |
| **Articles in Peer-reviewed Journals** | Chmielewski NN, Caressi C, Giedzinski E, Parihar VK, Limoli CL. "Contrasting the effects of proton irradiation on dendritic complexity of subiculum neurons in wild type and MCAT mice." Environ Mol Mutagen. 2016 Jun;57(5):364-71. Epub 2016 Mar 20. http://dx.doi.org/10.1002/em.22006 ; PubMed PMID: 26996825 , Jun-2016 |
| **Articles in Peer-reviewed Journals** | Acharya MM, Baddour AA, Kawashita T, Allen BD, Syage AR, Nguyen TH, Yoon N, Giedzinski E, Yu L, Parihar VK, Baulch JE. "Epigenetic determinants of space radiation-induced cognitive dysfunction." Sci Rep. 2017 Feb 21;7:42885. http://dx.doi.org/10.1038/srep42885 ; PubMed PMID: 28220892; PubMed Central PMCID: PMC5318883 , Feb-2017 |
| **Articles in Peer-reviewed Journals** | Lee SH, Dudok B, Parihar VK, Jung KM, Zöldi M, Kang YJ, Maroso M, Alexander AL, Nelson GA, Piomelli D, Katona I, Limoli CL, Soltesz I. "Neurophysiology of space travel: Energetic solar particles cause cell type-specific plasticity of neurotransmission." Brain Struct Funct. 2017 Jul;222(5):2345-57. Epub 2016 Nov 30. http://dx.doi.org/10.1007/s00429-016-1345-3 ; PubMed PMID: 27905022 ; PubMed Central PMCID: PMC5504243 , Jul-2017 |